SKILL BAY ACADEMY · KAUVERY HOSPITAL CCDP

# Google Gemini API — Full Project Integration Analysis & Master Roadmap

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|---|---|---|---|
| **Target System:** | AI Placement Analytics Platform (CCDP) | **Date:** | October 2026 |
| **Architecture:** | FastAPI + React 19/Vite + Google GenAI | **Audit Status:** | **Active & Verified** |
| **Current Model:** | gemini-3.6-flash (Discovered via models.list) | **SDK Version:** | google-genai (Official SDK) |

**Executive Summary:** This technical audit provides an exhaustive, code-level analysis of every location where the Google Gemini API is referenced, utilized, or required across both backend and frontend repositories. The project has an active, connected Gemini integration via the official google-genai SDK in llm_manager.py. However, the codebase exhibits **architectural fragmentation**: three legacy modules (ai_service.py, chat_service.py, schema_detector.py) continue to use raw, deprecated httpx REST calls with rigid API-key prefix guards (startswith('AIza')) that fail on newer Google keys. Furthermore, critical features—including on-demand student re-analysis, Excel batch narrative synthesis, and live executive report summaries—currently bypass Gemini with static placeholders. This specification outlines the exact remediation path to unify the entire application around a single, resilient GenAI core.

## 1. Current Gemini API Implementation Map

The following table catalogues every occurrence of Google Gemini API references, SDK invocations, and environment variable hooks in the active codebase. Each occurrence is evaluated for operational necessity.

| File Path | Lines | Code / Invocation | Functional Description | Necessity |
|---|---|---|---|---|
| **backend/app/services/llm_manager.py** | 24-26<br>63-71<br>503-542 | `genai.Client()`<br>`client.chats.create()`<br>`chat.send_message_stream()` | Official SDK client. Orchestrates streaming SSE, function-calling analytics tools, and multi-model fallback. | **CRITICAL (Core Engine)** |
| **backend/app/services/llm_manager.py** | 74-121 | `client.models.list()` | Dynamic model discovery at runtime. Filters retired models and caches active models for frontend dropdowns. | **REQUIRED** |
| **backend/app/routers/ai.py** | 54-65<br>120-168 | `/api/ai/status`<br>`/api/ai/models`<br>`/api/ai/chat/stream` | FastAPI router exposing status, active model inventory, and SSE streaming to MiniChatbot and SkillBay AI. | **REQUIRED (API Surface)** |
| **backend/app/services/ai_service.py** | 123-145<br>147-156 | `httpx.post(url)`<br>`generativelanguage/v1beta` | Generates single-student AI summaries, 30-day plans, and certifications. Contains buggy startswith('AIza') check. | **REDUNDANT (Refactor to SDK)** |
| **backend/app/services/chat_service.py** | 338-383 | `_gemini_grounded_chat()`<br>`httpx.post()` | Legacy RAG query responder with manual HTTPX calls. Orphaned by llm_manager.py tool calling. | **DUPLICATE (Deprecate)** |

| | | | | |
|---|---|---|---|---|
| **backend/ schema_detector.py** | 340-389 | `_ask_llm_for_schema()` `httpx.post()` | Calls LLM to infer table schema and score columns from Excel first 15 rows. Defaults to retired gemini-1.5-flash. | **REDUNDANT (Refactor to SDK)** |
| **backend/app/ config.py** | 15-24 | `gemini_api_key` `gemini_model_default` | Pydantic BaseSettings loading API keys, default models, timeout (20s), and model fallback sequence. | **CRITICAL (Config)** |
| **backend/app/routers/ admin.py** | 63-64 | `gemini_model` `gemini_api_key_configured` | Reports active Gemini model and configuration status to the admin settings dashboard. | **REQUIRED** |
| **frontend/src/hooks/ useAiChat.js** | 27-39 196-265 | `/api/ai/chat/stream` `SSE event parser` | Frontend client hook managing chat history, model selection, attachments, and real-time token streaming. | **CRITICAL (Frontend)** |
| **frontend/src/pages/ SkillBayAI.jsx** | 106-110 172-185 | `Model selector dropdown` `Session sidebar` | Dedicated full-screen AI chat page supporting model selection, session memory, and attachment parsing. | **REQUIRED** |
| **frontend/src/pages/ UploadStudentData.jsx** | 95, 117 814-830 | `reportModel` `EFFORT_LEVELS` | UI controls to select model and effort for instant PDF/Image export from uploaded session preview. | **REQUIRED (Unconnected)** |
| **frontend/src/pages/ StudentProfile.jsx** | 175-177 | `ai_source === 'gemini'` | Renders AI badge indicating whether student's narrative was generated via Gemini or local heuristics. | **REQUIRED** |

## 2. Architectural Fragmentation & Duplicate Implementations

The audit revealed three critical architectural bottlenecks that must be resolved to ensure production stability:

- **1. Dual API Calling Paradigms:** The system utilizes both the modern google-genai SDK (in llm_manager.py) and legacy direct httpx.post() REST requests (in ai_service.py, chat_service.py, and schema_detector.py). This causes maintenance fragmentation, conflicting timeouts, and inconsistent model resolution.
- **2. Fragile Key Prefix Validation:** Three legacy modules enforce api_key.startswith('AIza'). Google AI Studio now provisions valid production keys beginning with AQ.... Consequently, valid keys cause these modules to silently fail and drop into offline fallback, while llm_manager.py functions properly.
- **3. Retired Model Fallbacks:** References to gemini-1.5-flash persist in schema_detector.py, reports.py, and render.yaml. When invoked, these trigger HTTP 404 / 400 errors. All calls must route through resolve_working_model() with gemini-3.6-flash as default.
- **4. Orphaned Chat Engine:** chat_service.py contains an older grounded RAG implementation. The active router ai.py routes chat through llm_manager.py via tool calling. chat_service.py's LLM route is dead code and should be removed.

## 3. Required Gemini API Integration Locations

The following five key operational workflows currently lack Gemini integration or rely on static placeholders. Each requires immediate implementation to unlock full platform capabilities:

| Workflow / Target File | Insertion Line | Required Code & Logic | Business Justification | Connection Module |
|---|---|---|---|---|
| **A. Student Profile On-Demand Re-Analysis** backend/app/routers/students.py | After line 150 | `@router.post('/{id}/ai-analysis')` Calls `llm_manager.generate_student_narrative()` with model selection. | Allows placement officers to regenerate tailored 30-day plans and roadmaps directly from the student profile. | **app.services.llm_manager** |
| **B. Upload Pipeline Batch Narrative Synthesis** backend/app/routers/upload.py | Lines 107-112 in persist_consolidated_student | Replace hardcoded strings with async batch call or background task using `generate_ai_report()`. | Prevents uploaded cohorts from storing generic boilerplate roadmaps. Ensures unique Gemini evaluations. | **app.services.llm_manager** |
| **C. Live Cohort Executive Report Summary** backend/app/routers/reports.py | Lines 794-796 in _build_live_report_pdf | Query Gemini for a 3-sentence executive cohort diagnosis and embed it into the generated PDF story. | Delivers instant, high-value AI commentary on batch strengths and Kauvery role readiness for executive leadership. | **app.services.llm_manager** |
| **D. Intelligent Role Fit Explainer** backend/app/routers/jobs.py | After line 150 | `@router.post('/roles/{id}/ai-explain')` Takes student scores and returns clinical fit rationale. | Explains *why* a student is a 90% fit for a Kauvery unit and details remediation steps for gaps. | **app.services.llm_manager** |
| **E. Admin Model Switching & Connection Test** backend/app/routers/admin.py | Lines 86-137 | Persist `gemini_model` in DB/settings and add `POST /settings/test-ai` to verify keys. | Allows administrators to switch active Gemini models dynamically from the frontend settings screen. | **app.config.settings** **llm_manager** |

## 4. Recommended Unified GenAI Architecture

To achieve enterprise reliability, the platform should transition to a **Single Gateway Pattern** where all features consume Gemini via llm_manager.py:

```
[ FRONTEND LAYER ]
• MiniChatbot & SkillBayAI ■■■ SSE Streaming (/api/ai/chat/stream)
• Student Profile View ■■■ On-Demand Analysis (/api/students/{id}/ai-analysis)
• Upload Workflow ■■■ Live Session Export (/api/reports/live/pdf with AI summary)
• Settings Panel ■■■ Model Switching & Key Health Check (/api/admin/settings)

                                    ▼ (JSON REST / SSE)
[ BACKEND UNIFIED LLM GATEWAY (llm_manager.py) ]
• Client Management: Single genai.Client initialized from settings.gemini_api_key
• Dynamic Discovery: models.list() caches working models every 5 minutes
• Auto-Fallback Chain: gemini-3.6-flash ➔ gemini-3.5-flash-lite ➔ local offline
• Grounding Engine: First-class read-only tool calling (Analytics, Cohort, Student Profile)
• Task Functions: generate_chat_turn(), generate_student_narrative(), generate_cohort_narrative()

                                    ▼ (HTTPS REST)
[ GOOGLE GEMINI CLOUD (Developer API v1beta) ]
• Multimodal Vision (Inbound PDF/PNG) • Structured JSON • Tool Calling • Fast Reasoning
```

## 5. Environment Variable Requirements & Cloud Matrix

The following matrix outlines all environment variables required across development, Render backend, and Vercel frontend environments:

| Variable Name | Target Service | Production Value / Example | Purpose & Description |
|---|---|---|---|
| `GEMINI_API_KEY` | Backend (Render) | AIzaSy... or AQ... | Google AI Studio API key. Grants access to Gemini models. |
| `GEMINI_MODEL_DEFAULT` | Backend (Render) | gemini-3.6-flash | Primary model for chat, streaming, and student narrative generation. |
| `GEMINI_MODEL_FAST` | Backend (Render) | gemini-3.5-flash-lite | High-speed model used during high traffic or automated schema classification. |
| `GEMINI_MODEL_DEEP` | Backend (Render) | gemini-3.8-flash | Deep reasoning model for executive reports and comprehensive evaluations. |
| `GEMINI_FALLBACKS` | Backend (Render) | gemini-3.6-flash,gemini-3.5-flash | Comma-separated fallback chain invoked automatically on 503 or 404. |
| `GEMINI_TIMEOUT_SECONDS` | Backend (Render) | 20.0 | Maximum duration before falling back to local deterministic response. |
| `VITE_API_URL` | Frontend (Vercel) | https://skillbay-backend.onrender.com | Directs frontend Axios requests and SSE streams to the Render backend. |
| `VITE_DEV_MODE` | Frontend (Vercel) | false | Enforces production authentication and real AI server endpoints. |

## 6. Step-by-Step Implementation Roadmap

To achieve 100% unified Gemini integration with zero breaking changes, follow this execution sequence:

**Phase 1: Key Validation & Legacy Clean-up** — Remove startswith('AIza') checks from ai_service.py, chat_service.py, and schema_detector.py. Ensure all modules accept AQ... keys. Replace retired gemini-1.5-flash defaults with gemini-3.6-flash.

**Phase 2: Unified Narrative Helpers in llm_manager.py** — Add dedicated helper functions in llm_manager.py: generate_student_narrative() (returning structured summary, 30-day plan, roadmap, certs) and generate_cohort_executive_summary() (for live report PDFs).

**Phase 3: Student Profile On-Demand Re-Analysis** — Create POST /api/students/{id}/ai-analysis in students.py. In StudentProfile.jsx, add a 'Regenerate AI Analysis' button with a spinner and instant badge refresh.

**Phase 4: Live Cohort PDF AI Commentary** — Update _build_live_report_pdf() in reports.py to invoke generate_cohort_executive_summary() and inject a styled AI Narrative callout box above the student table.

**Phase 5: Job Role AI Fit Explainer** — Add POST /api/jobs/roles/{id}/ai-explain in jobs.py to explain candidate strengths and specific Kauvery unit requirements.

**Phase 6: Admin Settings Model Persistence** — Enable updating settings.gemini_model in admin.py and add an instant connection diagnostic test button in Settings.jsx.

## 7. Final Verification Checklist

| Check Item | Verification Command / Target | Status |
|---|---|---|
| **1. Gemini SDK Initialized** | `genai.Client(api_key=...)` instantiated cleanly | **VERIFIED (Live)** |
| **2. Status Endpoint Working** | `GET /api/ai/status` returns `mode: gemini` | **VERIFIED (200 OK)** |
| **3. Dynamic Model Discovery** | `GET /api/ai/models` lists available models | **VERIFIED (200 OK)** |
| **4. Real-time Streaming SSE** | `POST /api/ai/chat/stream` emits `tool_status` & `token` | **VERIFIED (Live)** |
| **5. Automatic Model Fallback** | Falls back to `gemini-3.5-flash-lite` on 503 or 404 | **VERIFIED (Unit Test)** |
| **6. Student Narrative Unification** | Refactor `ai_service.py` to consume `llm_manager` | **PENDING ACTION** |
| **7. Cohort Report AI Narrative** | Embed executive summary into Live PDF builder | **PENDING ACTION** |
| **8. Zero Retired Model Refs** | Purge `gemini-1.5-flash` from configs & doc templates | **PENDING ACTION** |